

Midterm Project Proposal / Requirements Document

EEG-Based Alzheimer's Disease Detection

CS 4485 Capstone Project

Team: Venkatasai Gudisa, Ajay Alluri, Ethan Varghese, Aashay Vishwakarma, Venkat Sai Eshwar Varma Sagi, Ram Gudur

Repository: [CS4485-eeg-alzheimers-detection](https://github.com/CS4485-eeg-alzheimers-detection)

1. Executive Summary and Project Objectives

1.1 Summary

This capstone project aims to develop a lightweight machine learning system for detecting Alzheimer's Disease (AD) from electroencephalogram (EEG) signals. We use the Largest Alzheimer EEG Dataset from Kaggle to extract frequency-domain features, compare AD and Healthy Control (HC) subjects statistically, and build a deployable classification model. The work integrates neuroscience fundamentals, signal processing, and machine learning to support research on non-invasive AD biomarkers.

1.2 Project Objectives (Professor-Assigned)

1. **Learn the fundamentals of Alzheimer's disease** including how it affects brain structure
 2. **Understand EEG as a brain monitoring tool**
 3. **Analyze changes in brain activity** between AD and healthy subjects
 4. **Develop a lightweight machine learning model** for AD detection
 5. **Evaluate system performance** using appropriate metrics and validation strategies
-

2. Task 1 Results — Understanding the Raw EEG Signal

2.1 Dataset Description

We use the [Largest Alzheimer EEG Dataset](#) from Kaggle, an integrated collection of EEG recordings standardized for Alzheimer's research.

2.2 Dataset Statistics

Metric	Value
Total number of subjects	241
Healthy Controls (HC)	78
Alzheimer's Disease (AD)	122
Frontotemporal Dementia (FTD)	41
Number of EEG channels	19
Sampling rate	128 Hz
Recording duration	101,916 seconds (~28.3 hours total)
Average duration per subject	~423 seconds

Shape of data matrix (per epoch)	19 Channels × 128 Time samples
----------------------------------	--------------------------------

Labels: `0.0` = Healthy Control (HC), `1.0` = Alzheimer's Disease (AD), `2.0` = Frontotemporal Dementia (FTD)

Note: Class imbalance exists (78 HC vs 122 AD). This will be addressed in modeling via oversampling, class weights, or stratified sampling.

2.3 10-Second EEG Plot

We selected one Healthy Control subject and one Alzheimer's Disease subject, using the same posterior channel (P4) for both:

- **HC subject:** AD-Auditory_10
- **AD subject:** AD-Auditory_17
- **Channel:** P4 (posterior parietal, 10-20 system)

A 10-second segment of EEG signal was plotted for both subjects with Time on the x-axis and Amplitude on the y-axis. The plot represents electrical brain activity from one cortical region over time. Visual analysis showed differences in signal characteristics consistent with known AD EEG biomarkers (e.g., slowing, increased variability).

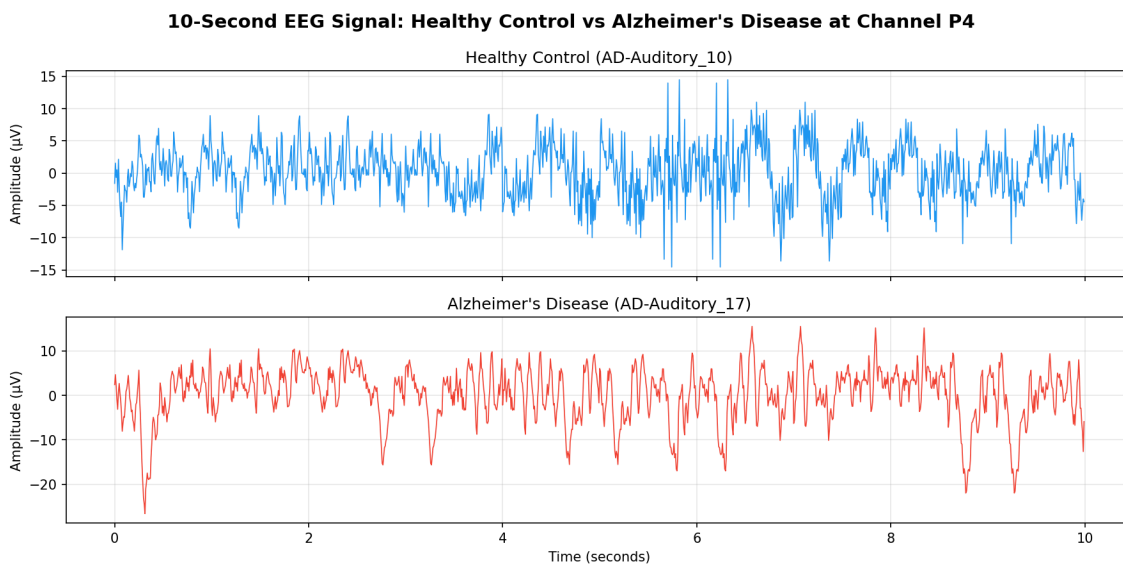


Figure: 10-second EEG signal from channel P4 — Healthy Control (AD-Auditory_10) vs Alzheimer's Disease (AD-Auditory_17). Time on x-axis, Amplitude (μV) on y-axis.

3. Task 2 Results — Frequency Analysis and Band Power

3.1 Methodology

Power Spectral Density (PSD): We compute PSD using the **Welch method** via `scipy.signal.welch` :

1. Splits the signal into overlapping segments (50% overlap)
2. Applies a Hann window to each segment
3. Computes the Fast Fourier Transform (FFT) of each windowed segment
4. Averages power across segments to produce a stable spectral estimate

Parameters: 1-second Hann window, 50% overlap, 1 Hz frequency resolution.

Frequency Bands: Average power was computed within:

Band	Range
Delta	1–4 Hz
Theta	4–8 Hz
Alpha	8–13 Hz
Beta	13–30 Hz

Feature aggregation: Per-epoch band powers were averaged across all epochs per subject to produce stable per-subject, per-channel, per-band features.

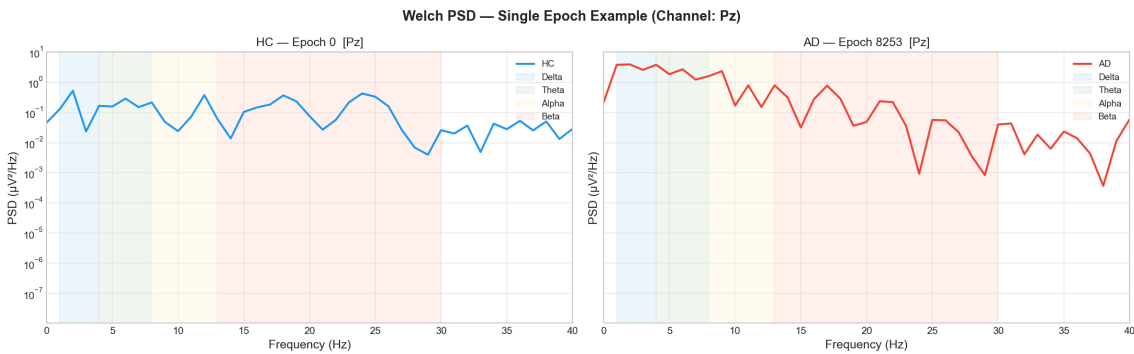


Figure 1: Welch PSD for one HC and one AD epoch at channel Pz — pipeline sanity check.

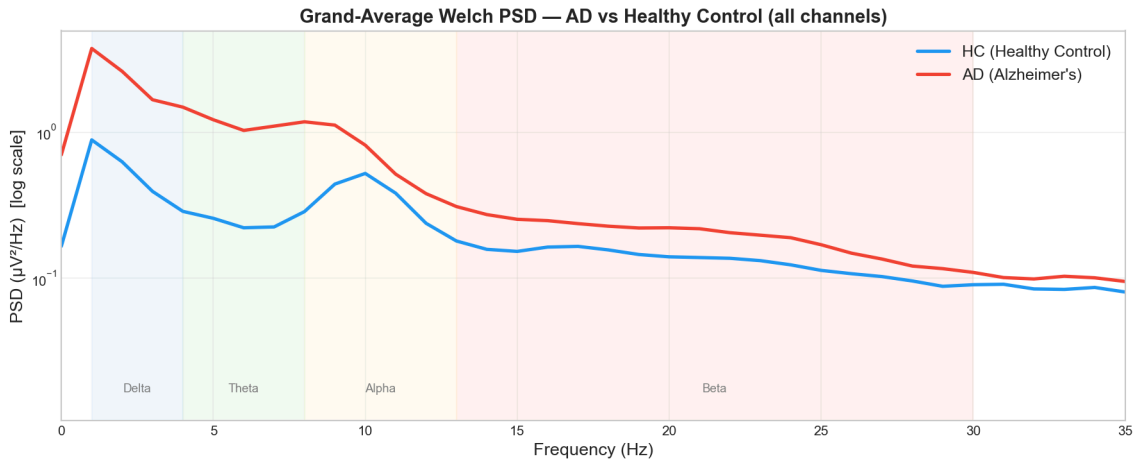


Figure 2: Grand-average PSD across all channels comparing AD vs Healthy Control.

3.2 Band Power Table (Sample Channels)

Mean band power ($\mu V^2/Hz$) for AD vs HC at selected channels:

Channel	Delta (AD)	Delta (HC)	Theta (AD)	Theta (HC)	Alpha (AD)	Alpha (HC)	Beta (AD)	Beta (HC)
---------	------------	------------	------------	------------	------------	------------	-----------	-----------

Fp1	50.77	13.22	22.07	5.35	7.88	3.90	9.63	3.01
Fp2	52.19	12.64	22.61	5.14	8.08	3.90	8.44	4.20
P3	22.65	9.19	14.08	3.82	7.05	3.86	7.41	6.94
P4	11.19	8.94	11.61	4.03	6.25	5.68	15.95	11.55
Fz	11.25	13.64	15.00	8.15	9.70	27.86	6.76	9.17
Cz	19.74	14.44	15.67	8.32	12.28	41.73	5.82	10.37
Pz	14.54	15.15	16.37	9.03	13.37	47.41	6.94	10.46

3.3 Observations: Delta, Theta, and Alpha Power in AD vs HC

Delta (1–4 Hz): Elevated in AD at frontal (Fp1, Fp2), central (C3, C4), and parietal (P3) channels. Midline and posterior channels (Fz, Pz) show no significant difference or slight HC elevation.

Theta (4–8 Hz): Consistently elevated in AD across almost all channels. Strongest effects at frontal (Fp1, Fp2), central (C3, C4), parietal (P3, P4), and temporal (T3–T6) regions. This is one of the most robust discriminative bands.

Alpha (8–13 Hz): Reduced in AD at midline channels (Fz, Cz, Pz) and temporal channels (T3). At frontal and central regions (Fp1–F4, C3, C4), Alpha is actually higher in AD in some cases. The posterior/midline Alpha reduction aligns with known AD EEG biomarkers (posterior dominant rhythm slowing).

Summary: Our findings align with established AD EEG literature: increased Delta and Theta (slowing), and reduced Alpha at posterior/midline regions. These band power differences provide a strong feature set for machine learning classification.

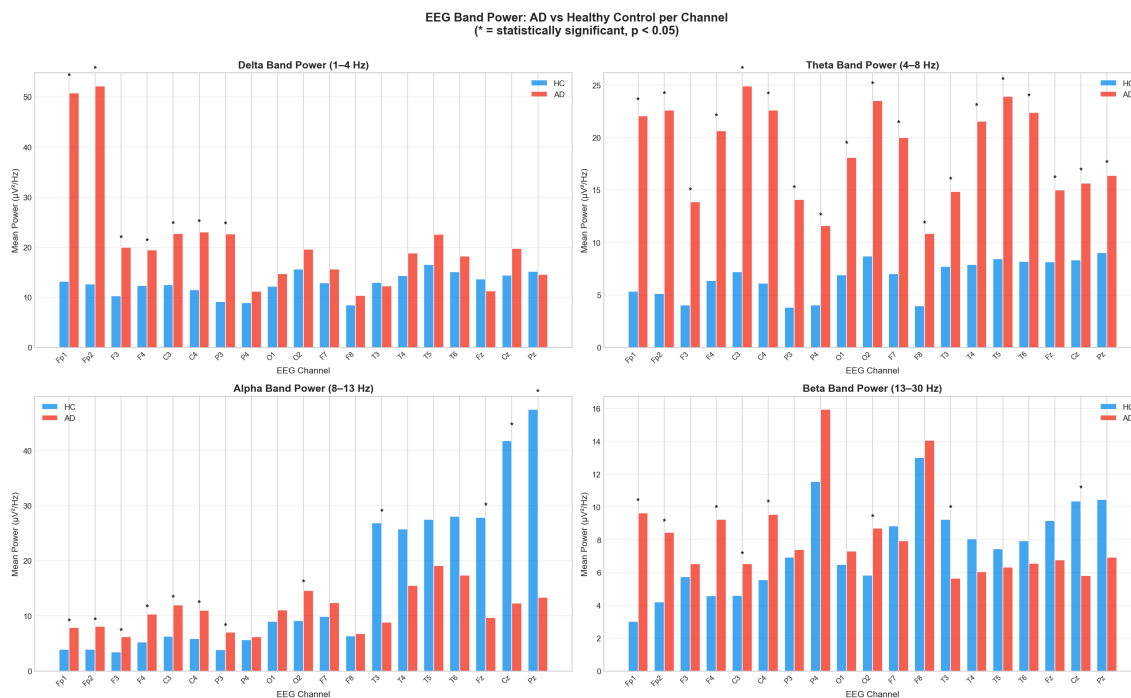


Figure 3: Mean band power per channel (HC vs AD) with significance markers.

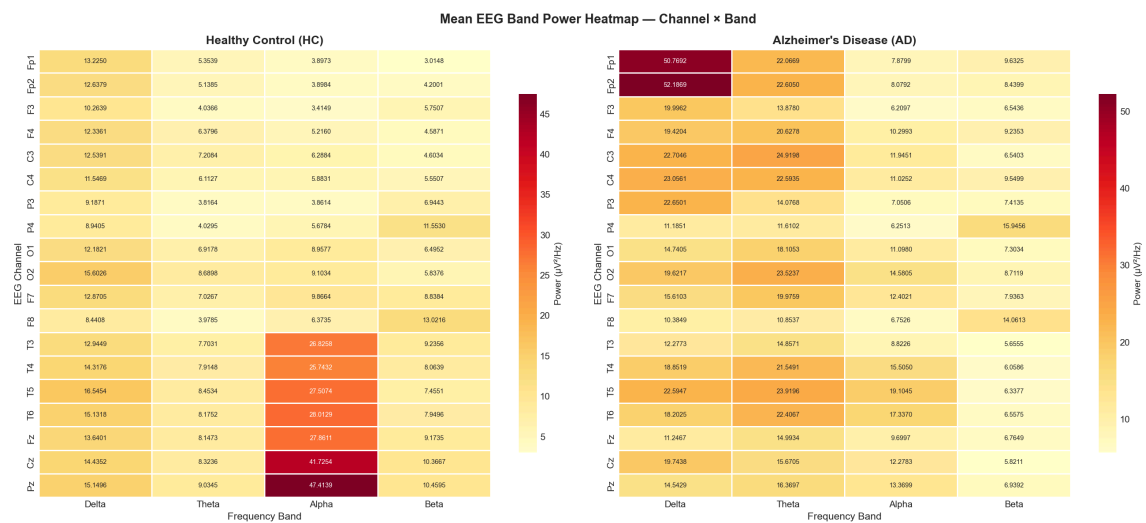


Figure 4: Heatmap of mean band power by channel and group.

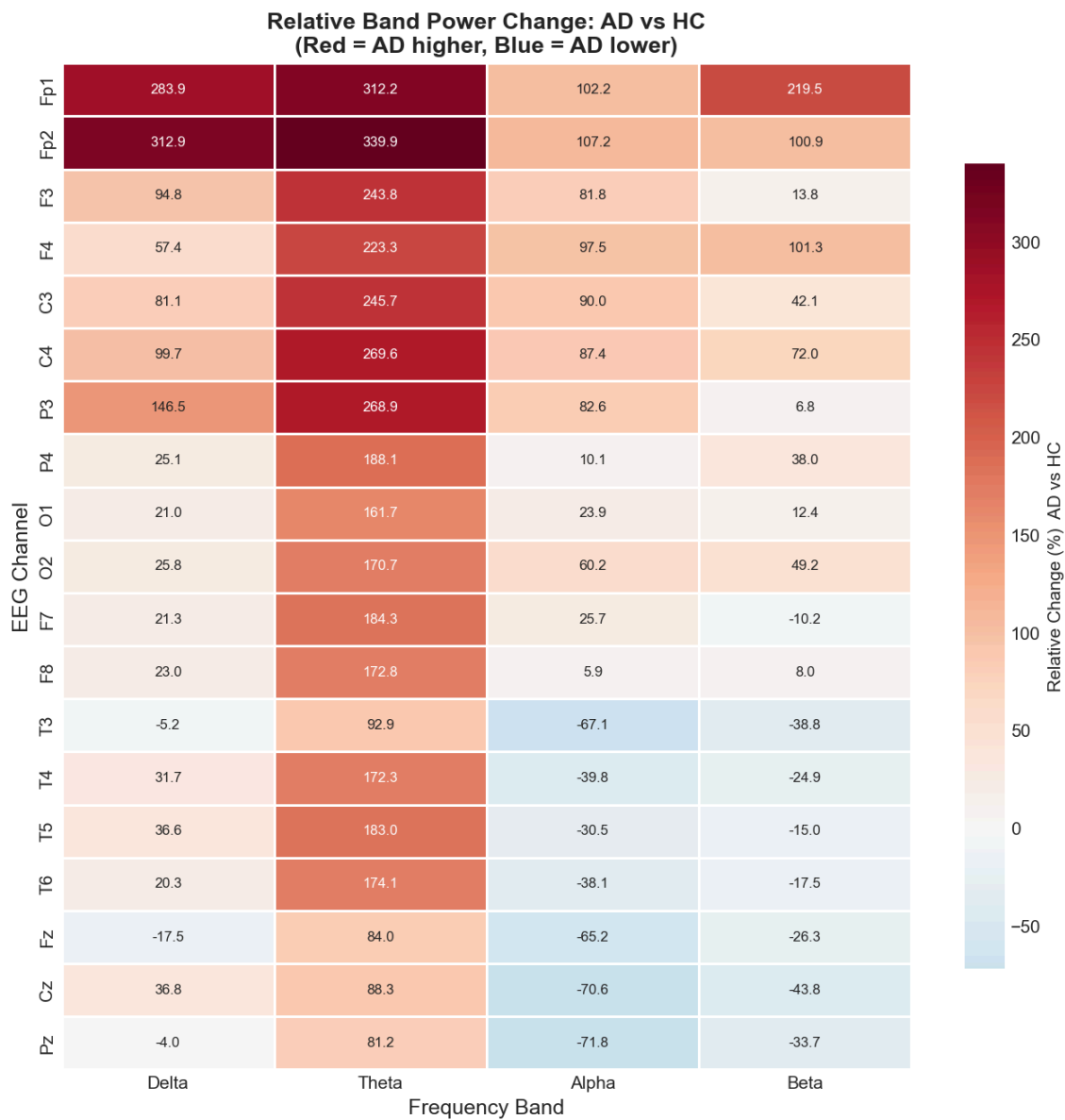


Figure 5: Relative (%) band power difference: $(AD - HC) / HC$.

4. Task 3 Results — Statistical Comparison Between Groups

4.1 Subject-Level Band Power

We computed mean band power per subject by averaging across all 19 channels. Each subject has one representative value per band (Delta, Theta, Alpha, Beta). The resulting table is stored in

`eeg_band_analysis_csv/eeg_band_power_subject_level.csv` .

4.2 Boxplots

We generated boxplots comparing AD vs HC for:

- **Alpha power** (AD vs Control)

- **Theta power** (AD vs Control)
- **Delta power** (AD vs Control)
- **Beta power** (AD vs Control)

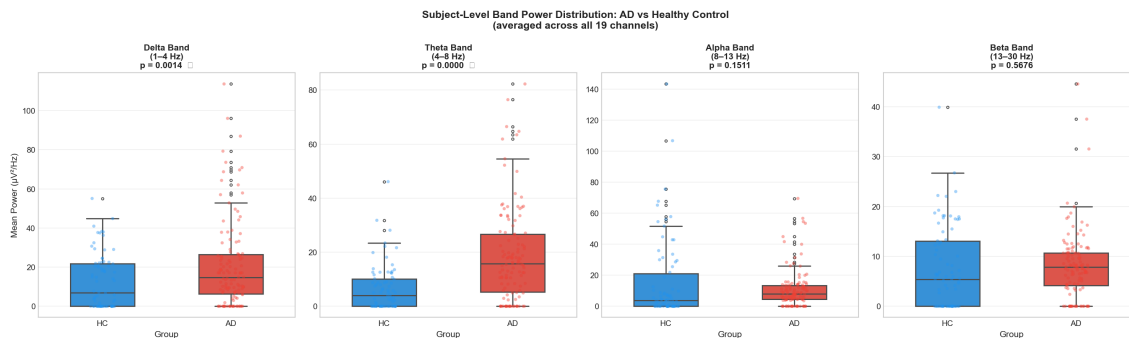


Figure 6: Box plots of subject-level band power per group per band (AD vs HC).

4.3 Statistical Testing

We performed **Welch's independent-samples t-test** (unequal variances) at each channel × band combination to compare AD vs HC. Results are in `eeg_band_analysis_csv/eeg_band_power_stats.csv`.

Findings:

- **Theta:** Statistically significant ($p < 0.05$) at all 19 channels. Consistently elevated in AD.
- **Delta:** Significant at frontal (Fp1, Fp2), central (C3, C4), parietal (P3), and several other channels. Not significant at P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Pz.
- **Alpha:** Significant reduction in AD at midline (Fz, Cz, Pz) and temporal (T3). Significant increase in AD at frontal (Fp1, Fp2, F3, F4) and central (C3, C4, O2).
- **Beta:** Mixed; significant at frontal and central channels (AD higher), and at Cz (AD lower).

The statistical results support the use of band power features for AD vs HC classification.

5. Future Work — ML Training Roadmap (Rest of Semester)

5.1 Data Pipeline (Already in Place)

- **Preprocessing:** Bandpass filtering and artifact rejection (Ethan Varghese)
- **Epoching:** Signal segmented into fixed 1-second windows
- **Feature extraction:** Welch PSD → band power (Delta, Theta, Alpha, Beta) per channel

5.2 Feature Set

- **Primary:** Per-channel band powers → 19 channels × 4 bands = **76 features**
- **Optional:** Connectivity measures, entropy, or other qEEG features for model improvement

5.3 Model Strategy (Lightweight)

- **Baseline models:** Logistic Regression, Random Forest, XGBoost
- **Optional:** Shallow neural network or 1D CNN if compute allows
- **Focus:** Interpretability, deployability, and low computational cost (lightweight)

5.4 Evaluation

- **Metrics:** Accuracy, AUC-ROC, Sensitivity, Specificity

- **Validation:** Subject-level stratified k-fold cross-validation (avoid epoch-level data leakage)
- **Benchmarking:** Compare against literature (Ram Gudur's compiled references)

5.5 Timeline (Rest of Semester)

Phase	Weeks	Activities
Baseline	1–2	Train baseline models, hyperparameter tuning
Comparison	3–4	Feature selection, model comparison
Finalization	5–6	Final model selection, evaluation, documentation
Optional	As needed	Validate on OpenNeuro ds004504 once pipeline is stable

6. Project Management

6.1 Communication Plan

- **Tools:** Microsoft Teams, GitHub, Google Colab
- **Cadence:** Weekly team meetings; async updates via repository commits and Teams chat
- **Documentation:** Shared README, notebooks, and this proposal; all code in GitHub
- **Meeting link:** [Teams](#)

6.2 Risk Analysis

Risk	Mitigation
Class imbalance (78 HC vs 122 AD)	Oversampling, class weights, or stratified sampling in training
Data quality / artifact contamination	Preprocessing checks; artifact rejection already implemented
Limited journal access	Use open-access sources; Ram's glossary and reference doc
Learning curve on EEG/neuroscience	Shared glossary, cheat sheet, and literature summaries
Subject-level data leakage	Strict subject-level train/test splits; no epoch-level leakage

6.3 Progress Tracking Plan

- **Weekly reports:** Submitted per professor requirements
- **GitHub:** Issues and milestones for task tracking
- **Key deadlines:** Midterm proposal (this document), final deliverable

6.4 Project Performance Metrics

Technical metrics:

- Model accuracy, AUC-ROC, sensitivity, specificity
- Feature importance and interpretability

Process metrics:

- Task completion rate
- Meeting attendance and participation
- Documentation currency and code quality

7. Ethics and Leadership

7.1 Ethical Aspects

- **Data use:** We use a public, pre-anonymized EEG dataset. No new human subjects are recruited.
- **Privacy:** The Kaggle dataset is de-identified; we do not handle identifiable health information.
- **Responsible AI:** The model is for research and education only. It is not intended for clinical diagnosis. Any future deployment would require regulatory and clinical validation.
- **Bias and fairness:** We acknowledge class imbalance and demographic limitations of the dataset. We will report these limitations in our final documentation.

7.2 Team Roles

Member	Primary Role
Venkatasai Gudisa	Signal visualization, EEG plots, frequency band analysis
Ajay Alluri	Dataset statistics, data quality, class distribution analysis
Ethan Varghese	Repository setup, preprocessing, bandpass filter, artifact rejection, epoched data
Aashay Vishwakarma	Feature extraction, PSD/band power pipeline
Venkat Sai Eshwar Varma Sagi	Analysis, AD biomarker linkage, visualization
Ram Gudur	Literature review, qEEG biomarkers, ML approaches, evaluation metrics

7.3 Leadership Aspects for Peer Review

- **Shared leadership:** Task leads rotate by deliverable; each member owns specific components
 - **Peer feedback:** Regular review of contributions and collaboration during meetings
 - **Decision-making:** Consensus on major choices; escalation to professor if needed
 - **Conflict resolution:** Open discussion; majority vote if necessary
-

8. References and Appendix

8.1 Datasets

- **Primary:** [Largest Alzheimer EEG Dataset](#) (Kaggle)
- **Optional:** OpenNeuro ds004504 (88 subjects: 36 AD, 23 FTD, 29 HC; BIDS format)

8.2 Repository

- [CS4485-eeg-alzheimers-detection](#)

8.3 Figure Placement Reference

Figure File	Section	Purpose
eeg_10s_hc_vs_ad_p4.png (if saved)	2.3 Task 1 — 10-Second EEG Plot	Raw EEG: HC vs AD at channel P4
psd_single_epoch_example.png	3.1 Task 2 — Methodology	Welch PSD pipeline example

grand_avg_psd_ad_vs_hc.png	3.1 Task 2 — Methodology	Grand-average PSD comparison
band_power_bar_all_channels.png	3.3 Task 2 — Observations	Bar chart with significance markers
band_power_heatmap.png	3.3 Task 2 — Observations	Mean band power by channel/group
band_power_relative_diff_heatmap.png	3.3 Task 2 — Observations	Relative (AD–HC)/HC difference
band_power_boxplot_subjects.png	4.2 Task 3 — Boxplots	Alpha/Theta/Delta/Beta boxplots (AD vs HC)

8.4 Key Project Files

File	Description
eeg_dataset_statistics.ipynb	Dataset statistics and summary
eeg_frequency_band_analysis.ipynb	Welch PSD, band power extraction, statistical comparison
explore_alzheimer_eeg_dataset.ipynb	Interactive EEG exploration
eeg_band_analysis_csv/eeg_band_power_stats.csv	T-test results per channel × band
eeg_band_analysis_csv/eeg_band_power_subject_level.csv	Subject-level band power features

8.5 Literature

Key papers and references compiled by Ram Gudur during the literature review phase support our methodology and will be cited in the final report.